



Maximilien Richer, Deuxfleurs

`https://garagehq.deuxfleurs.fr/ Matrix channel: #garage:deuxfleurs.fr`

Our objective at Deuxfleurs

French association promoting digital sovereignty and privacy
through self-hosting hosting **as an alternative to large cloud providers**

This requires resilience

(we want good uptime/availability with low supervision)

But what is Garage, exactly?

Garage is a self-hosted drop-in replacement for the Amazon S3 object store that implements resilience through geographical redundancy on commodity hardware



What makes Garage different?

Coordination-free:

- ▶ No Raft or Paxos
- ▶ Internal data types are CRDTs
- ▶ All nodes are equivalent (no master/leader/index node)

→ less sensitive to higher latencies between nodes

What makes Garage different?

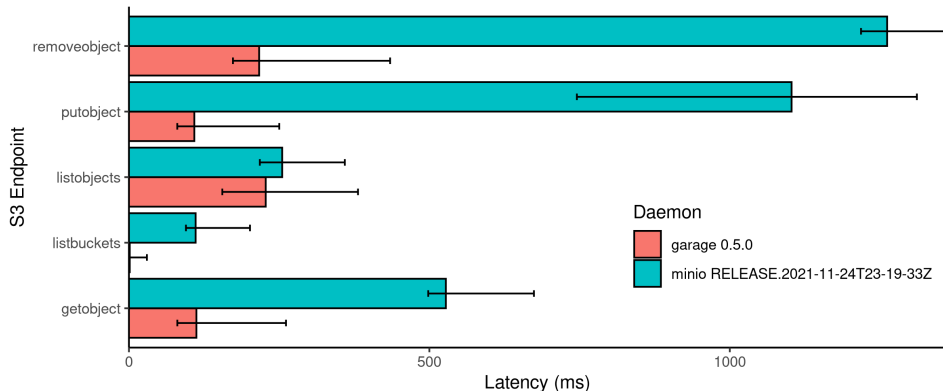
TODO update with latest garage and minio versions

S3 endpoint latency in a simulated geo-distributed cluster

100 measurements, 6 nodes in 3 DC (2 nodes/DC), 100ms RTT + 20ms jitter between DC

no contention: latency is due to intra-cluster communications

colored bar = mean latency, error bar = min and max latency



Get the code to reproduce this graph at <https://git.deuxfleurs.fr/quentin/benchmarks>

What makes Garage different?

Consistency model:

- ▶ Not ACID (not required by S3 spec) / not linearizable
- ▶ **Read-after-write consistency**
(stronger than eventual consistency)

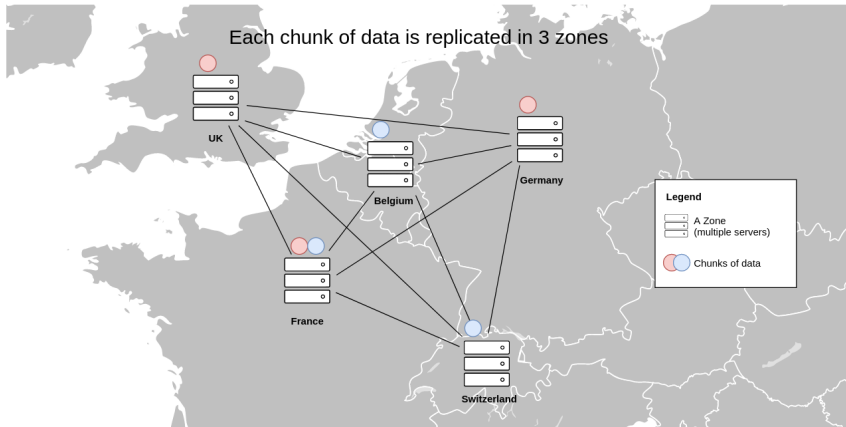
What makes Garage different?

Location-aware:

```
alex@io:~$ docker exec -ti garage /garage status
==== HEALTHY NODES ====
ID                Hostname  Address                               Tags                               Zone  Capacity
d9b5959e58a3ab8c... drosera   [2a01:e0a:260:b5b0::4]:3901        [drosera,atuin]                   atuin  20
156d0f7a88b1e328... digitale [2a01:e0a:260:b5b0::3]:3901        [digitale,atuin]                   atuin  10
966dfc7ed8049744... datura   [2a01:e0a:260:b5b0::2]:3901        [datura,atuin]                     atuin  10
7d50f042280fea98... io       [2a01:e0a:5e4:1d0::57]:3901        [io,jupiter]                       jupiter 20
alex@io:~$
```

Garage replicates data on different zones when possible

What makes Garage different?



An ever-increasing compatibility list



[matrix]



CyberDuck



R**CLONE**



PeerTube



Garage

Version history and roadmap

- ▶ v0.3: initial beta release (2021)
- ▶ v0.7: first released version (2022)
- ▶ v1.0: stable release (2024), will be deprecated in summer 2026 1y after v2.0 was released
- ▶ v2.0: stable release (2025)
 - ▶ new HTTP admin API
 - ▶ reworted replication configuration: `replication_mode` changed to `replication_factor` & `consistency_policy`
- ▶
v3.0: TBA may include versionning support, tag on buckets and objets, retention policies...

Best practices for Garage deployments

Things you should know

- ▶ no TLS support, use your own proxy
- ▶ no anonymous access (use website endpoint)
- ▶ you need to assign roles to nodes manually
- ▶ the replication factor cannot be changed easily
- ▶ the default region is garage and not us-east-1
- ▶ only use the degraded consistency policy for data recovery!

What hardware should I use?

- ▶ do NOT use network file storage (NFS, SMB, etc.) for metadata
- ▶ get a **write-intensive flash disk** for the metadata folder
- ▶ set it up as a RAID1 if possible, with a COW filesystem (e.g. Btrfs or ZFS)
- ▶ get large HDDs for the data folder
- ▶ use XFS and garage multi-hdd mode for best performance
- ▶ you can use a RAID for data but you'll leave a lot of performance on the table

Garage doesn't require a powerful CPUs nor much RAM, but your performance will depend on your disks!

Picking a metadata engine

All files-to-block mappings are stored in the metadata engine, including bucket and object metadata. Files below 3KB are stored directly in the metadata engine.

- ▶ Sled: removed in 1.x, move to SQLite or LMDB
- ▶ **SQLite**: safer, **recommended for small clusters and single-node**
- ▶ LMDB: faster, recommended for large clusters with metadata redundancy
 - ▶ Warning: limited to 480 bytes per key with LMDB (not an issue in practice)
- ▶ Fjall: experimental but promising rust-native engine, test it and let us know!

Metadata engine can be set node per node, and changed later with a migration tool

Single-node deployment

- ▶ garage was initially designed for multi-node deployments
- ▶ single-node deployments are possible, but you will lose resilience
- ▶ **If you do please ensure you have backups** (especially for metadata)
 - ▶ set up `metadata_auto_snapshot_interval`
- ▶ use sqlite to minimize data loss risks on powercuts
- ▶ or use a UPS!

Use github.com/bikeshedder/garage-single-node for an easy single-node setup!

Multi-node deployment

- ▶ try to have geo-distributed zones
- ▶ multiple nodes per zone to add more capacity
- ▶ at least 3 zones for best resilience
- ▶ keep in mind your available network and IO bandwidth
- ▶ **Rebalancing a cluster can take multiple weeks with large HDDs and slow network links**
- ▶ monitor your nodes with Prometheus + Grafana

Deuxfleurs has been running a 9TB (3TB usable) 8-nodes cluster (3+3+2) over retail fiber (10ms site-to-site latency) for close to 5 years now. We heard there are petabyte clusters out there!

Deploying and administering garage at scale

- ▶ deploy with your favorite tool (eg. Ansible) and system manager (eg. systemd)
- ▶ or use Docker, docker-compose, Kubernetes or Nomad
- ▶ Kubernetes and Consul are supported for node-to-node discovery
 - ▶ you'll still have to manage the layout manually!
- ▶ use gateway nodes to optimize network usage
- ▶ adjust `resync-tranquility` and `scrub-tranquility` to your resources

Kubernetes storage controller:

`github.com/bmarinov/garage-storage-controller`

Community UI available!



Garage

WebUI



Dashboard



Cluster



Buckets



Keys

Dashboard



Healthy
Status



5
Nodes



5
Connected Nodes



5
Storage Nodes



5
Active Storage Nodes



256
Partitions



256
Partitions Quorum



256
Active Partitions



91.7 MB
Total Usage

Embedded UI comming this year!

 Garage

Web Admin

Overview

STORAGE

Buckets

ACCESS

Access Keys

Admin Tokens

CLUSTER

Nodes

Layout

Connected node **garage-n1** 

Last updated 2min ago  Refresh

Overview

Connected nodes

5 / 5 known

Storages nodes

4 / 4  Cluster Healthy

100% connected

Data space available

671 GB estimated

Metadata space available

663 GB estimated

Buckets

450 active

→

Access keys

5 / 10 active

→

Admin tokens

3 / 9 active

→

Storage health

All nodes →

 garage-n1

Resync queue: 2255
Resync error: 1 
Scrub corruption: 0

Data Disk usage: 45%
Metadata Disk usage: 30%

225 GB free
30 GB free

→

 garage-n2
last seen 2min ago

Resync queue: 2255
Resync error: 0
Scrub corruption: 0

Disk usage: 45%

225 GB free

→

 garage-n3

Resync queue: 2255
Resync error: 1 
Scrub corruption: 0

Disk usage: 45%

225 GB free

→

Embedded UI comming this year!

 Garage
Web Admin

Overview

STORAGE

Buckets

ACCESS

Access Keys

Admin Tokens

CLUSTER

Nodes

Layout

Buckets / my-bucket / Keys allowed

Connected node garage-n1 Help

my-bucket

bucket-alias local , super-bucket-alias local [Go manage aliases ->](#)

7ac5a87a176f4eba003ea49d6db5aea5fa42eef21bb19c2a51d44968d30ca7f9

Last updated 2min ago

Refresh

Actions

Creation date	Space	Objects	Keys	Unfinished Uploads	Website access
22 Feb 2024	3 GB 50% of 6 GB	64 065 12% of 500K	8 1 expired	565 (300 MB) with 6436 parts	Enabled

Keys

Aliases

Website access

Keys allowed

+ Allow a key

7 displayed

Sort

Search for name / id

Name / id	Creation date	Permissions	Expiration	Aliases	
a-key-name GKb74ad322b45899a20d4872b8	22 Feb 2025	Read/Write	∞	1	
a-key-name GKb74ad322b45899a20d4872b8	13 Dec 2024	Read/Write/Owner	13 Dec 2024 expired	0	
a-key-name GKb74ad322b45899a20d4872b8	22 Nov 2024	Read/Write	∞	2	

Demo time!

Get Garage now!



Garage

<https://garagehq.deuxfleurs.fr/>
Matrix channel: #garage:deuxfleurs.fr

